

systemd Timers Cheat Sheet

Run jobs on a schedule with systemd: a .timer file says **when**, a matching .service file says **what**.

Why use a timer instead of cron

You get	How	You get	How
Output in the journal	journalctl -u backup ; no lost mail	Catch-up after downtime	Persistent=true (runs at boot)
No overlapping runs	Won't start again while still running	Order and limits	After= , MemoryMax= etc.

backup.service: what to run

```
# /etc/systemd/system/backup.service
[Unit]
Description=Nightly backup
[Service] ← no [Install]: the timer starts it
Type=oneshot ← runs to the end, then exits
ExecStart=/usr/local/bin/backup.sh ← full path
User=backup ← optional; default root
```

backup.timer: when to run it

```
# /etc/systemd/system/backup.timer
[Unit]
Description=Run backup nightly
[Timer] ← same name = runs backup.service
OnCalendar=*-*-* 02:30:00 ← every day at 02:30
Persistent=true ← missed while off? run at boot
RandomizedDelaySec=10min ← random 0-10 min delay
[Install]
WantedBy=timers.target ← needed for enable
```

When: OnCalendar= and friends

Value	Fires	Value	Fires
daily	Every day at 00:00	Mon..Fri 09:00	Weekdays at 9 am
weekly	Mondays at 00:00	*:0/15	Every 15 minutes (:00, :15, :30, :45)
monthly	1st of the month, 00:00	*-*-* 02:30:00	Every day at 02:30 (year-month-day time)
quarterly	1 Jan, Apr, Jul, Oct, 00:00	Sat *-*-1..7 03:00	First Saturday of each month
OnBootSec=5min	5 min after boot	OnUnitActiveSec=1h	1 h after it last ran (use with OnBootSec)

```
$ systemd-analyze calendar 'Mon..Fri 09:00' ← test first · times are local unless you add a zone (09:00 UTC)
Original form: Mon..Fri 09:00 ← what you typed
Normalized form: Mon..Fri *-*- * 09:00:00 ← how systemd reads it
Next elapse: Mon 2026-10-05 09:00:00 UTC ← next time it would fire
From now: 1 day 21h left
```

Turn it on and check it

```
$ sudo systemctl daemon-reload ← after creating or editing either file
$ sudo systemctl enable --now backup.timer ← the .timer, not the .service
$ systemctl list-timers ← --all includes inactive timers
```

NEXT	LEFT	LAST	PASSED	UNIT	ACTIVATES
Sun 2026-10-04 00:00:00 UTC	11h left	Sat 2026-10-03 00:00:04 UTC	12h ago	logrotate.timer	logrotate.service
Sun 2026-10-04 02:30:00 UTC	14h left	Sat 2026-10-03 02:37:41 UTC	9h ago	backup.timer	backup.service

NEXT/LEFT: next run, and how soon. **LAST/PASSED**: when it last ran. **UNIT** → **ACTIVATES**: timer → service it starts.

Handy recipes

Command	Does
sudo systemctl start backup.service	Runs the job right now, to test it
journalctl -u backup.service -n 50	Output of the last runs
sudo systemd-run --on-calendar='2026-10-04 02:00' /root/fix.sh	One-off timer, no unit files to write
systemctl --user enable --now sync.timer	Units in ~/.config/systemd/user/
sudo loginctl enable-linger mark	mark's timers run when logged out

Remember: enable the .timer, not the .service. **systemctl list-timers** shows what's scheduled; **journalctl -u name.service** shows what happened when it ran.